CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT

Course Code and Title**: S-CSPC315 - ALGORITHMS AND COMPLEXITY**

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

DIRECTIONS: Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, As you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

Format, parts, and description	Compliance	
	Complied With	Not Complied With
Format		
1. The 2023-2024 syllabus format is used.	/	
2. The font used in the syllabus preparation is Times New Roman (11).	/	
3. DLSUD (being first) and college logos are beside each other at the top (center).	/	
4. Preliminaries include all the contact details of the professor, such as the day and time.	/	
5. Each WEEK has a corresponding date.	/	
Content		
6. The Learning Outcomes have more HOTS than LOTS.	/	
7. CLOs and TLOs are identified for each module/lesson/topic.	/	
8. Gospel reading and Reflection and reflection precede the presentation of each topic under the teaching-learning activities.	/	
9. The journal articles/references are within the last five years.	/	
10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies.	/	
11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students.	/	
Assessments		
12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus.	/	
13. Formative Assessments are not reflected anymore in the syllabus.	/	
14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities.	/	
15. The syllabus presents one-module(lesson)-one-assessment correspondence.	/	
16. Only one Enabling Assessment is reflected in a given week	/	
17. The Enabling Assessments are aligned with the identified CLOs and TLOs.	/	
18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term.	/	
19. Thirty percent (30%) is allotted for Summative Assessments per term.	/	
20. Twenty percent (20%) is allotted for Class Participation/Attendance per term.	/	

21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade.	/	
22. Summative Assessments are aligned with the identified CLOs, TLOs and teaching-learning activities.	/	
23. Enabling Assessments are aligned with the identified CLOs and TLOs and teaching-learning activities.	/	
24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD	/	
25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD.	/	
References and Curriculum Map		
26. The References must be within the last 5 years except for canons and seminal studies.	/	
27. It must be a combination of books, eBooks, journal articles, videos, etc.	/	
28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS.	/	
29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs)	/	

NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

Self-reviewed:

Tita Ramos Herradura

Checked, verified, and endorsed:



Rolando B. Barrameda

Dept. Curriculum Committee Reviewer

Checked, verified, and approved:

Josephine T. Eduardo ***Dr. Rubie M. Causaren***

Department Chair College Dean

De La Salle University - Dasmariñas
COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-CSPC315
COURSE TITLE	**:**	ALGORITHMS AND COMPLEXITY
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	3 Units
LEARNING MODALITY:		Traditional Blended Learning Model
PRE-REQUISITES	**:**	S-ITCS214, S-ITCS214LA
CO-REQUISITES	**:**	None
PROFESSOR	**:**	TITA RAMOS HERRADURA
CONTACT DETAILS	**:**	trherradura@dlsud.edu.ph /SB (M-F, 8:00AM – 5:00 PM)

COURSE DESCRIPTION:

This course covers the design and analysis of different algorithms under various design principles and techniques making use of different data structures. It presents the relevance of algorithm in everyday lives and in computing science. Through algorithmic complexity measures, different range of behaviors of algorithms will be understood. It includes several analysis and design techniques such as divide-and-conquer, dynamic programming, greedy algorithms, and others. This course aims to enable students, who will be future computer scientist, to create correct and efficient algorithms and programs.

The course will have a special provision on modality, pacing, pedagogy, assessment type, communication and feedback based on the national, regional, provincial, city, and institutional guidelines for safety amidst disaster and pandemic. Hence, this class shall be delivered in both online and offline/module-based approach. This course is also compliant with Outcomes-Based Education (OBE) and 21st Century Learning Design (21CLD) frameworks.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:
CLO1. exhibit the knowledge of relevance of algorithm in computing science;
CLO2. understand the different algorithm design principles and techniques;
CLO3. analyze the design of algorithms and their complexities;
CLO4. construct effective and efficient algorithms for different problems and scenarios using various principles and techniques;
CLO5. apply advanced algorithm techniques using any programming language to a software development project; and
CLO6. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Exhibit the knowledge of relevance of algorithm in computing science.	TLO1. Identify the criteria of an algorithm. TLO2. Understand the importance of algorithm analysis
CLO2. Understand the different algorithm design principles and techniques.	TLO3. Be familiar with the different algorithm design and analysis.
CLO3. Analyze the design of algorithms and their complexities.	TLO4. techniques Analyze the efficiency of the algorithm. TLO5. Compute the running time complexity of a given program segment.
CLO4. Construct effective and efficient algorithms for different problems and scenarios using various principles and techniques.	TLO6. Compare the performance of different algorithms solving similar tasks.
CLO5. Apply advanced algorithm techniques using any programming language to a software development project.	TLO7. Convert algorithm into a running computer program.
CLO6. Demonstrate 21st century skills in all learning activities.	*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Week**	**Teaching-Learning Activities / Assessment Strategy** **Module Activities**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Summative**	**Self-care**	**Allotted Hours**
Module 1 CLO 6	1 8/14-8/27	• Gospel reading and reflection • Course syllabus uploaded in SB • Course overview • Course requirement details	Schoolbook PowerPoint	/				3.0
Module 1 TLO 1 TLO 2 CLO 1 CLO 6	2 8/21-8/26	• Gospel reading and reflection • Warm-up • Introduction to algorithm • Exercises (write a simple algorithm) • Enabling assessment: Quiz on algorithm definition and criteria	Schoolbook PowerPoint		/			3.0
Module 2 TLO 3 TLO 4	3 8/28-9/2	• Gospel reading and reflection • Discussion of running time complexity • Enabling assessment	Schoolbook Powerpoint	/				3.0
Module 3 TLO 3 TLO 4 CLO 3 CLO 4 CLO 6	4 -6 9/4-9/23	• Gospel reading and reflection • Analysis and Design Techniques: Recursive Routines (Iterations, Factorial, Fibonacci) • Lesson exercises- programming	Schoolbook Powerpoint Online IDE	/	/			9.0

Module 4 TLO 3 TLO 4 CLO 3 CLO 4 CLO 6	7 9/25-9/30	• Gospel reading and reflection • Analysis and Design Techniques: Divide & Conquer (mergesort, quicksort, binary search) • Lesson exercises	Schoolbook MS Stream	/				3.0
	8 10/2-10/7	• Gospel reading and reflection • Analysis and Design Techniques: Divide & Conquer (mergesort, quicksort, binary search) • Lesson exercises	Schoolbook Sway		/			3.0
	9 10/9-10/14	• SUMMATIVE ASSESSMENT	Schoolbook			/		3.0
			TOTAL	***12.0***	***12.0***	***3.0***	***0.0***	***27.0***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO3. Analyze the design of algorithms and their complexities.	TLO4. techniques Analyze the efficiency of the algorithm. TLO5. Compute the running time complexity of a given program segment.
CLO4. Construct effective and efficient algorithms and programs for different problems and scenarios using various principles and techniques.	TLO6. Compare the performance of different algorithms solving similar tasks.
CLO5. Apply advanced algorithm techniques using any programming language to a software development project.	TLO7. Convert algorithm into a running computer program. TLO8. Create a document about the advanced algorithm techniques used, description, features and the programming codes.
CLO6. Demonstrate 21st century skills in all learning activities.	*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Week**	**Teaching-Learning Activities / Assessment Strategy**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Summative**	**Self-care**	**Allotted Hours**
Module 5 TLO 6 TLO 7 CLO 3 CLO 4 CLO 6		**Module Activities**						
	10 10/16-10/21	• Gospel reading and reflection • Final output comprehensive instruction • Advanced design and analysis technique (Dynamic Programming- Fibonacci number series, 0/1 knapsack problem) • Lesson exercises	Schoolbook Sway		/			3.0
	11 10/23-10/28	SELF-CARE / ME-TIME						3.0
	12 10/30-11/4	• Gospel reading and reflection • Advanced design and analysis technique (Dynamic Programming- Fibonacci number series, 0/1 knapsack problem) • Enabling assessment: programming exercises	Schoolbook		/			3.0

Module 6 TLO 6 TLO 7 CLO 3 CLO 4 CLO 6	13 11/6-11/11	• Gospel reading and reflection • Advanced design and analysis technique (Greedy algorithms) Enabling assessment: programming exercises	Schoolbook	/				3.0
	14 11/13-11/18	• Gospel reading and reflection • Advanced design and analysis technique (Greedy algorithms)	Schoolbook	/				3.0
Module 7 TLO 7 TLO 8 CLO 3 CLO 4 CLO 5 CLO 6	15 -16 11/20-12/2	• Gospel reading and reflection • Advanced design and analysis technique (Backtracking Algorithms)	Schoolbook MS Stream	/	/			3.0
	17 12/4-12/9	• Final Output Completion	Schoolbook					3.0
	18 12/11-12/16	• SUMMATIVE ASSESSMENT	Schoolbook		/			3.0
			TOTAL	***9.0***	***12.0***	***3.0***	***3.0***	***27.0***

GRADING SYSTEM:

Midterm

Attendance/Class Participation	- 20%
Enabling Assessments	- 50%
Summative Assessments	- 30%
	100%

Finals

Attendance/Class Participation	- 20%
Enabling Assessments	- 50%
Summative Assessments	- 30%
	100%

Overall Final Grade = $\frac{\text{Midterm + Final}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes are delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.

ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
   b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones also cannot be used as calculator during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

On-Site References

Call number or e-provider	Reference Material
https://eds.s.ebscohost.com/eds/detail/detail?vid=8&sid=0f489988-6f33-4a61-bcda-ceebd658c668%40redis&bdata=JkF1dGhUeXBlPXNoaWImc2l0ZT1lZHMtbGl2ZQ%3d%3d#AN=2706813&db=e000xww	Chopra, R., & Raheja, S. (2020). Design and Analysis of Algorithms. NEW AGE International Publishers.
https://eds.s.ebscohost.com/eds/detail/detail?vid=1&sid=8a24813d-244a-4ae5-adbb-1344870b2548%40redis&bdata=JkF1dGhUeXBlPXNoaWImc2l0ZT1lZHMtbGl2ZQ%3d%3d#AN=18	Giuseppe Bonaccorso. (2018). Mastering Machine Learning Algorithms: Expert Techniques to Implement Popular Machine Learning Algorithms and Fine-tune Your Models. Packt Publishing.

23677&db=e000xww	

Online References

Call number or e-provider	Reference Material
Online	Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to algorithms. MIT press. https://dl.ebooksworld.ir/books/Introduction.to.Algorithms.4th.Leiserson.Stein.Rivest.Cormen.MIT.Press.9780262046305.EBooksWorld.ir.pdf
Online	GeeksforGeeks. A Computer Science Portal for Geeks. https://www.geeksforgeeks.org/fundamentals-of-algorithms/

Prepared: **Computer Science Department**
1st Semester 2023-2024

Approved: **Josephine T. Eduardo**
Chair, CSD

Endorsed: **Rubie M. Causaren, PhD**
Dean, CSCS

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COLLEGE OF SCIENCE AND COMPUTER STUDIES

COMPUTER SCIENCE DEPARTMENT

PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

COURSE CODE : S-CSPC315
COURSE TITLE : ALGORITHMS AND COMPLEXITY
COURSE TYPE : Lecture
COURSE CREDIT : 3 Units

Course Learning Outcome	Program Learning Outcome							
	PLO 1	**PLO 2**	**PLO 3**	**PLO 4**	**PLO 5**	**PLO 6**	**PLO 7**	**PLO 8**
1. Exhibit the knowledge of relevance of algorithm in computing science.	X		X		X			
2. Understand the different algorithm design principles and techniques.		X	X	X				
3. Analyze the design of algorithms and their complexities.		X	X		X			
4. Construct effective and efficient algorithms for different problems and scenarios using various principles and techniques.	X	X	X	X	X		X	X
5. Apply advanced algorithm techniques using any programming language to a software development project.		X	X		X			
6. Demonstrate 21st century skills in all learning activities.				X	X		X	X

NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)

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COMMON GE LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

COURSE CODE : S-CSPC315
COURSE TITLE : ALGORITHMS AND COMPLEXITY
COURSE TYPE : Lecture
COURSE CREDIT : 3 Units

Course Learning Outcome	Common GE Learning Outcome				
	CGE1	CGE2	CGE3	CGE4	CGE5
1. Exhibit the knowledge of relevance of algorithm in computing science.	X			X	X
2. Understand the different algorithm design principles and techniques.	X			X	X
3. Analyze the design of algorithms and their complexities.				X	X
4. Construct effective and efficient algorithms for different problems and scenarios using various principles and techniques.				X	X
5. Apply advanced algorithm techniques using any programming language to a software development project.					
6. Demonstrate 21st century skills in all learning activities.	X		X	X	X

NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)

COMMON GE LEARNING OUTCOME

1. Critiques/evaluates individual's character, actions, and decisions based on Christian-like perspective
2. Assesses situations and proposes ways that benefit the country, particularly its environment and its marginalized sector
3. Explains in various occasions the relationship of the course/subject with the other subjects within or outside the discipline
4. Recommends ways to improve the subject/course

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GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) – PROGRAM LEARNING OUTCOME

MAPPING TABLE FOR BS COMPUTER SCIENCE

Program Learning Outcome	Graduate Attributes/Institutional Learning Outcomes				
	GA1	GA2	GA3	GA4	GA5
1. Apply concepts of computing and mathematics in different domains of computer science.				X	X
2. Identify and analyze problems and define computing requirements appropriate to their solutions.					X
3. Design, develop, test and evaluate various solutions using algorithms for complex computing problems that take full advantage of usability of computer resources.			X	X	X
4. Document computer-based systems by writing effective reports for risk- assessment impact.				X	X
5. Work effectively as an individual or as a member of a team during software development.			X	X	X
6. Communicate effectively with the computing community and non-computer science professionals about computing activities.				X	X
7. Recognize the legal, social, ethical and professional issues involved in the utilization of computer technology and address these issues.	X	X	X		
8. Engage in continuing professional development through performing independent or collaborative research as computing professionals.			X	X	X

NOTE: Provide a check mark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expect from its graduates.

Graduate Attribute (GA)	**Descriptors (Institutional Learning Outcome)[1]**
1. God-centered	a) Creates an environment where the experience of God is lived and shared b) Practices honesty, fairness, truth and integrity in all aspects of life (personal and professional lives) c) Observes and maintains ethical standards in dealing with the different stakeholders d) Integrates Christian perspectives and values in all undertakings e) Manifests humility and respect in relating with other people
2. Patriotic	a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts c) Engages actively in political, social, economic, and cultural transformation for nation building d) Brings pride and honor to the community and the country e) Patronizes locally produced products and promotes them globally
3. Socially Responsive	a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation b) Engages in ecological advocacies as responsible stewards of God's creation c) Continuously works in solidarity with people and institutions to effect liberating action d) Brings forth awareness and deeper understanding of social realities e) Manifests spirit of volunteerism through social actions that improve communities
4. Globally Competitive	a) Takes progressive responsibility for own learning and development b) Promotes critical and creative thinking, self-knowledge and self-mastery c) Is committed to rise above mediocrity d) Shows dedication, commitment and love for work to attain exceptional results e) Exhibits innovativeness and creativity in various contexts
5. Professionally Competent	a) Possesses leadership qualities of **C**ompetence, **H**onesty, **A**ccountability, **R**esponsibility and **T**ransparency (CHART) b) Effectively communicates in various forms of media c) Strives to be at the top of their chosen fields d) Effectively works and collaborates in a multi-disciplinary team e) Adapts to changes in the workplace, local society and global community